HW 5

1. Show that, if it exists, the LU decomposition of $A \in \mathbb{R}^{n\times n}$ is unique if L is unit lower triangular.

Let L_1, L_2 be unit lower triangular and U_1, U_2 be upper triangular such that $A = L_1U_1$ and $A = L_2U_2$ where

$$A = L_1U_1 = L_2U_2$$
$$= \underbrace{L_2^{-1}L_1}_{\tilde{L}} = \underbrace{U_2U_1^{-1}}_{\tilde{U}}$$

We can verify that $\tilde{L} = L_2^{-1}L_1$ is unit lower triangular because multiplying 2 unit lower triangular matrices is also unit lower triangular. $\tilde{U} = U_2U_1^{-1}$ is upper triangular because multiplying 2 upper triangular matrices is also upper triangular. We may also use the fact that the only way that an upper triangular matrix equals an upper triangular is if they are both the same diagonal matrix. Since $\tilde{L}$ is unit lower triangular, they must be the identity,

$$L_2^{-1}L_1 = I \quad \Rightarrow \quad L_1 = L_2$$

$$U_2U_1^{-1} = I \quad \Rightarrow \quad U_2 = U_1$$

Thus, the LU decomposition is unique when L is unit lower triangular.

2. A is a symmetric positive definite (SPD) matrix

a) Prove the diagonal entries of A are strictly greater than 0

A matrix is positive definite if, for every nonzero vector $x \in \mathbb{R}^n$,

$$x^T A x > 0$$

Since we are only interested in the diagonal of A, we can use the basis vectors $e_1, \dots, e_n$ of $\mathbb{R}^n$ to isolate the a_{ii} entry. Since e_i is the standard basis vector with a 1 at the ith entry and is nonzero, we can apply the positive definite property,

$$e_i^T A e_i = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij} e_i e_j = \begin{cases} a_{ii}, & i=j \\ 0, & \text{otherwise} \end{cases}$$

Where this multiplication gets us entry a_{ii} since all other terms cancel out when $i \neq j$ with the only intersection at a_{ii}. Thus,

$$e_i^T A e_i = a_{ii} > 0 \quad \text{for} \quad 0 \leq i \leq n$$

so, the diagonal must be all strictly greater than 0.

3. Compute a Cholesky factorization of

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 7 \end{pmatrix} = \begin{pmatrix} l_{11} & & \\ l_{21} & l_{22} & \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} & l_{31} \\ & l_{22} & l_{32} \\ & & l_{33} \end{pmatrix}$$

Before computing $A = LL^T$, we must first verify A is SPD.

① Symmetric: $A = A^T$ ✓

② Positive definite: x^TAx Key: factor into LDU, check if diagonal of D is >0

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 7 \end{pmatrix} \underset{R_3 - R_1}{\overset{R_2 - R_1}{=}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 6 \end{pmatrix} \overset{R_3 - R_2}{=} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}$$

Pivots: $(1, 1, 5)$ are all >0

↳ A is SPD

$l_{11} = \sqrt{a_{11}} = \sqrt{1} = 1$ $\quad l_{22} = \left(a_{22} - \frac{a_{21}^2}{a_{11}}\right)^{1/2} = \sqrt{2 - \frac{1^2}{1}} = \sqrt{1} = 1$

$l_{21} = \frac{a_{21}}{\sqrt{a_{11}}} = \frac{1}{\sqrt{1}} = 1$ $\quad l_{32} = \left(a_{32} - l_{31}l_{21}\right)^{1/2} / l_{22} = \sqrt{2 - 1 \cdot 1} / 1 = 1$

$l_{31} = \frac{a_{31}}{\sqrt{a_{11}}} = \frac{1}{\sqrt{1}} = 1$ $\quad a_{33} = l_{31}^2 + l_{32}^2 + l_{33}^2 \Rightarrow l_{33} = \left(a_{33} - l_{31}^2 - l_{32}^2\right)^{1/2} = \sqrt{7 - 1^2 - 1^2} = \sqrt{5}$

$$\boxed{A = \underbrace{\begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 1 & \sqrt{5} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ & 1 & 1 \\ & & \sqrt{5} \end{bmatrix}}_{L^T}}$$

4. Let $A \in \mathbb{R}^{n\times n}$ be SPD

a) Verify $\langle x,y\rangle_A = y^TAx$ is $\langle \cdot,\cdot\rangle$ on $\mathbb{R}^n$

① Positivity: $\langle x,x\rangle > 0$ for all $x > 0$

$\langle x,x\rangle_A = x^TAx > 0$ because A is positive definite

② Symmetry: $\langle x,y\rangle_A = \langle y,x\rangle_A$

Since y^TAx is a 1×1 scalar, it is equal to its tranpose x^TAy

$\langle x,y\rangle_A = y^TAx = (y^TAx)^T = x^TAy = \langle y,x\rangle_A$

③ Humogeneity: $\langle \alpha x,y\rangle_A = \alpha\langle x,y\rangle_A$

$\langle \alpha x, y\rangle_A = y^TA(\alpha x) = \alpha y^TAx = \alpha\langle x,y\rangle_A$

since scalar multiplication is associative

④ Linearity: $\langle x+z,y\rangle_A = \langle x,y\rangle_A + \langle z,y\rangle_A$

$\langle x+z,y\rangle_A = y^TA(x+z) = y^TAx + y^TAz = \langle x,y\rangle_A + \langle z,y\rangle_A$

b) $\|x\|_A = \langle x,x\rangle_A^{1/2}$. Prove $|A(i,j)| \le \sqrt{A(i,i)A(j,j)}$ for $i \ne j$

Use Cauchy-Schwarz!

$|\langle x,y\rangle_A^T| \le \|x\|_A\|y\|_A \Rightarrow |y^TAx| \le \sqrt{x^TAx}\sqrt{y^TAy}$

Let e_i and e_j be the standard basis vectors of A with a 1 at row i and row j respectively

$e_j^TAe_i = A(i,j)$

$|y^TAx| = |e_j^TAe_i| \le \sqrt{e_i^TAe_i}\sqrt{e_j^TAe_j} = \sqrt{A(i,i)A(j,j)}$ ✓

5. Compute the orthogonal complement of the space in $\mathbb{R}^3$ spanned by $(1,-2,1)$

From class, the orthogonal complement of subspace $W = \text{span}\{(1,-2,1)\} \subseteq \mathbb{R}^3$ is

$$W^\perp = \{v \in \mathbb{R}^3 : \langle v, w \rangle = 0 \text{ for all } w \in W\}$$

Using the standard dot product, a vector v is orthogonal to the spanning vector $(1,-2,1)$ if $v \cdot (1,-2,1) = 0$ or

$$\begin{bmatrix} x & y & z \end{bmatrix}\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = x - 2y + z = 0 \Rightarrow x = 2y - z \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2y - z \\ y \\ z \end{pmatrix} = y\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Basis: $\boxed{\text{span}\left\{\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right\}}$ which is the orthogonal complement of $(1,-2,1)$

6. $A = \begin{bmatrix} L_{11} & & & \\ L_{21} & L_{22} & & \\ \vdots & \vdots & \ddots & \\ L_{n1} & L_{n2} & & L_{nn} \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & \cdots & U_{1n} \\ & U_{22} & \cdots & U_{2n} \\ & & \ddots & \vdots \\ & & & U_{nn} \end{bmatrix}$

flop count:

$$A_{ij} = \sum_{k=1}^{\min(i,j)} L_{i,k} U_{k,j}$$

$m = \min(i,j)$

$$\text{total flops} = \sum_{i=1}^{n}\sum_{j=1}^{n} \underbrace{(m + m-1)} = \sum_{i=1}^{n}\sum_{j=1}^{n} 2\min(i,j) - 1 = \begin{cases} i \geq j & \Rightarrow \min(i,j) = j \\ j > i & \Rightarrow \min(i,j) = i \end{cases} \Rightarrow 2\int_0^n \int_0^n \min(x,y)\,dy\,dx$$

m multiplications from dot product

m-1 additions from dot product

$$= 2\left(\frac{1}{3}n^3\right) = \frac{2}{3}n^3$$

This is 3x faster than matrix multiplication.

```python
import numpy as np


def compute_lu_product(L, U):
    n = L.shape[0]
    A = np.zeros((n, n))

    for i in range(n):
        for j in range(n):
            limit = min(i, j) + 1
            A[i, j] = np.dot(L[i, :limit], U[:limit, j])

    return A


L = np.array([[1, 0, 0], [0.5, 1, 0], [0.25, 0.75, 1]])
U = np.array([[4, 2, -1], [0, 1, 0.5], [0, 0, 2]])
print(f"L:\n{L}\n")
print(f"U:\n{U}\n")
A = compute_lu_product(L, U)
print(f"A:\n{A}")

''' TERMINAL OUTPUT
L:
[[1.   0.   0.  ]
 [0.5  1.   0.  ]
 [0.25 0.75 1.  ]]

U:
[[ 4.   2.  -1. ]
 [ 0.   1.   0.5]
 [ 0.   0.   2. ]]

A:
[[ 4.     2.    -1.   ]
 [ 2.     2.     0.   ]
 [ 1.     1.25   2.125]]
'''
```